



DVN36104 SPR 2026

Data Visualisation and Narratives
Spring 2026
Gerard Kelly





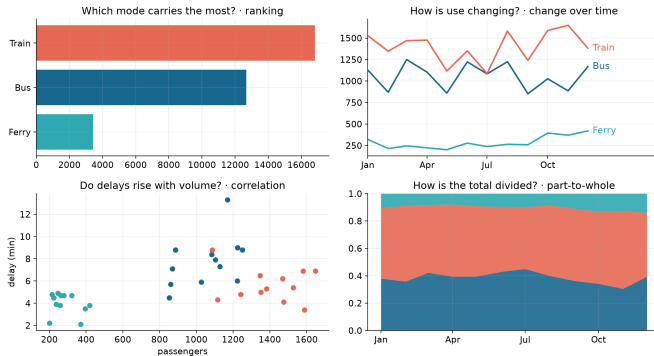
36104 · DATA VISUALISATION AND NARRATIVES

Choosing Visual Forms

Data types, encoding, tidy data, and the Visual Vocabulary

What visual task are we performing?

One dataset



audience → question → task → data shape → form — not data → favourite chart

Synthetic transport table: monthly bus, train and ferry passengers and delays, 2025 — the dataset you tidy in the notebook.

Goals

1. Classify fields by type and role.
2. Rank visual channels by precision.
3. Turn questions into tasks and forms.
4. Choose palettes from data semantics.
5. Reshape wide data to tidy, and verify.
6. Complete a Chart Choice Decision Record.

Data types

Type / role	Legitimate reading	Example
nominal	same or different; no order	transport mode · region
ordinal	higher or lower; spacing unknown	Mohs rank · service tier
quantitative, interval	differences meaningful; zero arbitrary	temperature in °C
quantitative, ratio	differences, ratios, true zero	passengers · dollars
temporal	sequence, interval, duration	month · timestamp
spatial / relational	location, connection, transfer	coordinates · links

A type can **rule out** an invalid comparison. It cannot select a chart without an audience and a question.

Bertin's variables

Bertin's visual variables

Position						
Size						
Value						
Texture						
Colour						
Orientation						
Shape						

Seven ways a mark can vary: position, size, value, texture, colour, orientation, shape.

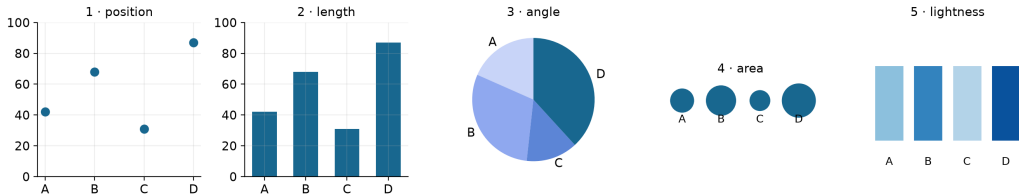
Bertin's question for each variable:

- is it **selective** — can the eye isolate a group?
- is it **ordered** — does it imply a sequence?
- is it **quantitative** — can it carry an amount?

After Bertin, *Sémiologie graphique* (1967). Course rendering.

Channel precision

The same four values, five channels — try reading B – A in each



Position on a common scale → length → angle/slope → area → colour lightness. The hierarchy is a design aid, not a law detached from audience and task.

After Cleveland & McGill (1984).

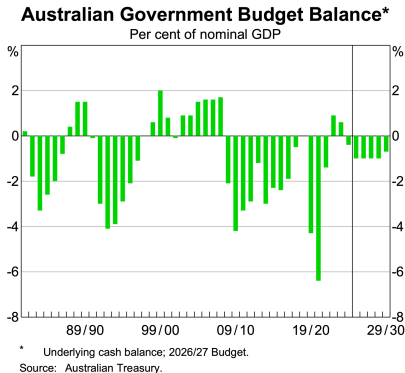


THE VISUAL VOCABULARY

Nine task families.

The originals on the slides.
You rebuild the modern forms in the notebook.

Deviation



FIELDS temporal or categorical + a signed measure with a meaningful zero

COMPARISON variation above and below a fixed reference — zero, a target or an average

FORMS diverging bar · surplus/deficit line · spine

Surplus above zero, deficit below. The notebook builds this family from the FRED yield spread.

Reserve Bank of Australia chart pack, Australian Government Budget Balance (% of

GDP) · source: Australian Treasury · © RBA, reproduced with attribution.

Correlation

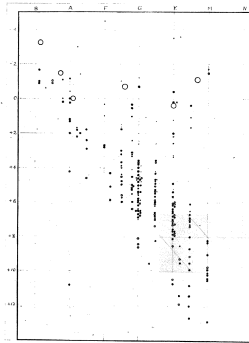


FIGURE 1.
This is a fairly tolerant criterion for a "good parallax", and the small

FIELDS two numeric variables, or
one numeric + one ordered proxy

COMPARISON direction, form,
strength, clusters, outliers

FORMS scatterplot · connected
scatter · bubble

Note the reversed magnitude axis: bright is
up.

H. N. Russell, Popular Astronomy 22 (1914), Figure 1 · scan courtesy Maria Mitchell

Observatory / NASA ADS · public domain.

Ranking

FIELDS categorical identifier + ordinal rank (+ optional true magnitude)

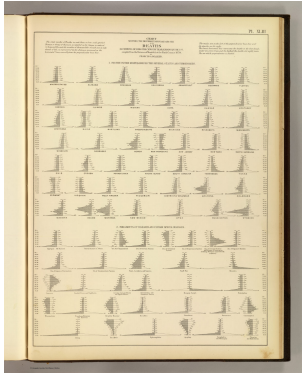
COMPARISON highest, lowest, relative ordered position

FORMS sorted bar · sorted dot · bump chart

Mohs published his 1812 scale as a ranked list — no chart existed. Ordinal rank hides that one step near the top is a five-fold jump in true hardness.

No historical chart survives; you draw this family in the notebook.

Distribution



F. A. Walker, 1874 age-and-sex distribution of deaths · Library of Congress · public

domain.

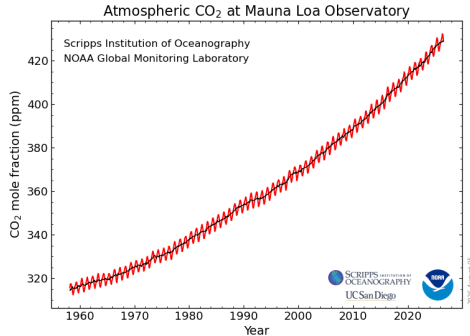
FIELDS one numeric variable;
optional group (sex) and ordinal bands
(age)

COMPARISON range, centre,
spread, shape, outliers

FORMS histogram · pyramid · box ·
violin

The 1874 plate shows deaths, not
population.

Change over time



FIELDS temporal variable + numeric measure; optional category

COMPARISON trend, rate of change, seasonality, turning points

FORMS line · slopegraph · area · small multiples

The window is part of the claim: crop it and the story changes.

Scripps Institution of Oceanography + NOAA Global Monitoring Laboratory · 5 Aug

2026.

Magnitude

FIELDS ordered categories + ratio quantity spanning orders of magnitude

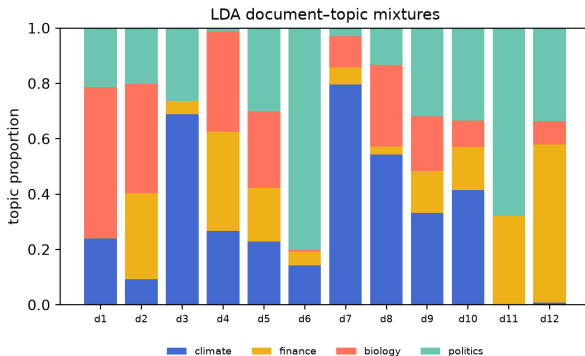
COMPARISON absolute amount on a common (often log) scale

FORMS bar · dot · lollipop; area is less precise

After Elton (1927) and Lindeman (1942). It looks like part-to-whole — it is not: ~90% of each level's energy is lost at the transfer, so the tiers sum to nothing.

No historical chart survives; you draw this family in the notebook.

Part-to-whole



Document-topic mixtures after Blei, Ng & Jordan (2003). Course rendering.

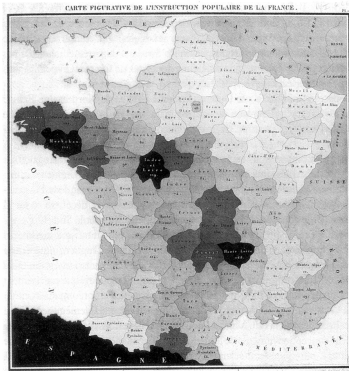
FIELDS document identifier + topic proportions summing to one

COMPARISON composition within each document and differences between documents

FORMS 100% stacked bar · waffle · treemap

Each bar is one document. Its topic shares sum to 100%.

Spatial



FIELDS geometry or location +
numeric or categorical attribute

COMPARISON regional pattern,
proximity, clustering

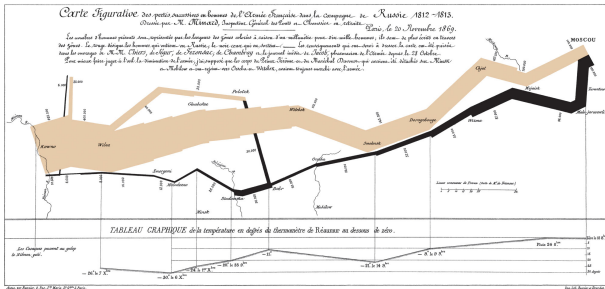
FORMS choropleth for rates ·
proportional symbol for counts · dot
map

The first modern choropleth. Map rates, not
raw counts.

C. Dupin, Carte figurative de l'instruction populaire de la France (1826) · BnF · public

domain.

Flow



C. J. Minard, Carte figurative...(1869) · public domain.

FIELDS source + target; optional weight and ordered stage

COMPARISON movement, connection, transfer, attrition

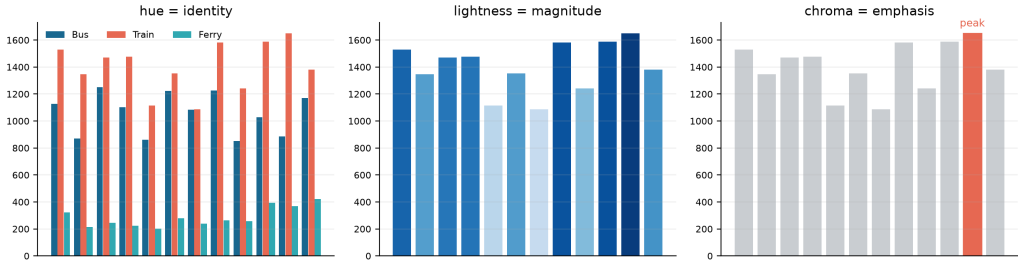
FORMS Sankey · alluvial · network · route map

Six variables held in one frame.

Field combinations

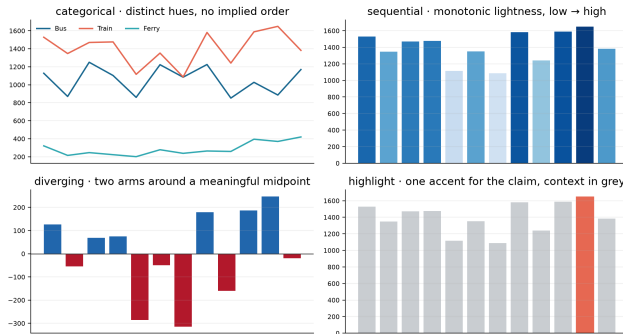
Fields	Required comparison	Task family	Common forms
1 numeric	shape, centre, spread	distribution	histogram · dot · box
1 categorical + 1 numeric	absolute size	magnitude	bar · dot · lollipop
1 categorical + 1 numeric	ordered position	ranking	sorted bar · sorted dot
1 temporal + 1 numeric	trend, rate of change	change over time	line · slope · area
2 numeric	paired association	correlation	scatterplot
1 numeric + 1 reference	signed difference	deviation	diverging bar · residual panel
categories + one total	share of whole	part-to-whole	100% bar · ring · treemap
1 spatial + 1 value	regional pattern	spatial	choropleth · prop. symbol
source + target (+ weight)	movement, connection	flow	Sankey · alluvial · route map

Colour



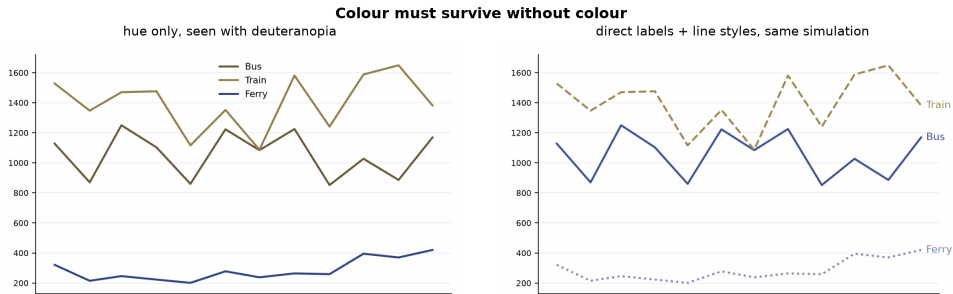
Colour is contextual and imprecise: let position or length carry exact values, and give colour the job it is actually good at.

Palettes



The diverging midpoint belongs to the **meaning of the measure** — zero, a target, a baseline — never to the palette designer.

Accessibility



Add redundant cues: direct labels, line style, shape, position. Test the actual chart, not the palette swatch. Distinguish missing from zero. Avoid rainbow ramps whose lightness invents boundaries.

Tidy data

Each variable a column · each observation a row · each value a cell.

Wickham (2014)

region	2023	2024	2025	→	region	year	riders
Inner	1.30M	1.41M	1.52M		Inner	2023	1.30M
Western	1.05M	1.13M	1.24M		Inner	2024	1.41M
					Western	2023	1.05M

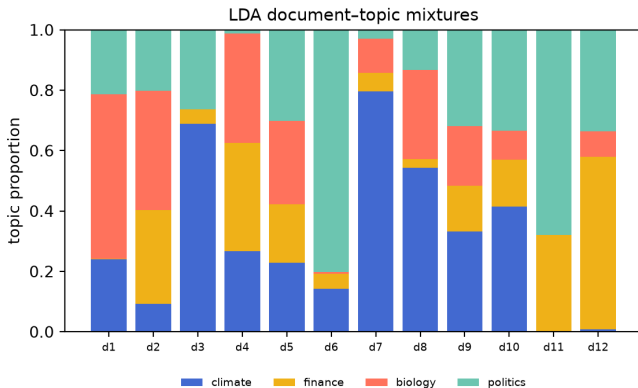
The class dataset arrives **deliberately messy**: a ‘.’ sentinel, a thousands separator, a missing value. None may be cleaned silently.

Verification

1. **Identifiers** — key columns intact; no invented or dropped categories.
2. **Row count** — does the reshaped count equal what arithmetic predicts?
3. **Types** — numbers numeric, dates dates, no silent strings.
4. **Missing** — where did NaNs appear, and are they real or created?
5. **Spot totals** — one hand-computed total matches the transformed table.

The notebook encodes each check as an `assert` you must satisfy.

Notebook Part 1



`choosing_visual_forms.ipynb`

For each family:

1. run the reconstruction;
2. satisfy the verification cell;
3. make one **consequential design intervention** and explain what it changes.

Simulated and representative inputs are labelled; do not cite them as observations.

Notebook Part 2

data types → field combinations → required comparison → task family → form

- Name the audience, question and decision **before** naming a chart.
- Reshape the messy wide table to tidy long form; pass the five checks.
- Build three candidates answering **different questions** — not three styles of one chart.
- Select for your claim; reject one plausible alternative for a specific reason.

A1 Part B

1. **Claim** — state a claim the supplied data can support.
2. **Task** — name the audience, question and required comparison.
3. **Alternative** — design a view that communicates that claim.
4. **Compare** — explain what becomes easier and harder to see than in the original.

The full Chart Choice Decision Record adds the Vocabulary category, the supporting field types, and a **genuine rejected alternative** with a specific reason. “It looked worse” is not sufficient.

Exit ticket

- My audience needs to answer ...
- This is primarily a _____ visual task.
- I selected _____ because ...
- I rejected _____ because ...

Then complete the short open-book diagnostic in Canvas after class. It is ungraded and completed without an assistant.

Credits

- Reserve Bank of Australia, chart pack: Australian Government Budget Balance — source Australian Treasury; © RBA, reproduced with attribution.
- H. N. Russell, *Popular Astronomy* 22 (1914), Figure 1 — scan courtesy Maria Mitchell Observatory / NASA ADS; public domain.
- F. A. Walker, age-and-sex distribution of deaths (1874) — Library of Congress, public domain.
- Scripps Institution of Oceanography + NOAA GML, Mauna Loa CO₂ record, 5 Aug 2026.
- D. Blei, A. Ng & M. Jordan, latent Dirichlet allocation (2003).
- C. Dupin (1826) — BnF, public domain. C. J. Minard (1869) — public domain.
- Course renderings: Bertin's variables, HR diagram, Mohs scale, population pyramid, Keeling curve, trophic pyramid, topic mixtures, choropleth and Minard reconstruction.
- Reading: FT Visual Vocabulary · Franconeri et al., *The Science of Visual Data Communication* · Wickham, *Tidy Data* · Cleveland & McGill (1984) · Bertin, *Sémiologie graphique* (1967).